# 使用 REST API 管理 ONTAP

ONTAP Automation

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# 使用 REST API 管理 ONTAP

## 使用 ONTAP REST API 管理 EMS 事件通知

您可以使用 ONTAP REST API 配置向您发送事件管理系统（ EMS ）事件通知的重要性，以及检索所有 EMS 消息或具有特定属性的特定消息。您可以使用 REST API 帮助自动执行与 EMS 相关的任务，与其他处理 EMS 消息的方法相比，可以节省时间。

以下示例显示了如何使用 ONTAP REST API 处理 EMS 消息。在每个示例中，请务必在所示位置用方括号 <> 提供信息值。您也可以引用示例 "Python 脚本" 这展示了如何自动化执行许多与 EMS 相关的活动。

### 查看特定事件日志

使用 `GET /support/EMS/Events` API 调用，您可以检索特定事件消息，例如最新消息，包含特定文本的消息或具有特定严重性的消息。

示例请求：检索最新消息

```
curl -X GET -u admin:<PASSWORD> -k
'https://<IP_ADDRESS>/api/support/ems/events?fields=message.name&max_recor
ds=1'
```

响应示例


```
"records": [
    {
      "node": {
        "name": "malha-vsim1",
        "uuid": "da4f9e62-9de3-11ec-976a-005056b369de",
        "_links": {
          "self": {
            "href": "/api/cluster/nodes/da4f9e62-9de3-11ec-976a-
005056b369de"
          }
        }
      },
      "index": 6260,
      "message": {
        "name": "netinet6.rtr.high.mtu"
      },
      "_links": {
        "self": {
          "href": "/api/support/ems/events/malha-vsim1/6260"
        }
      }
    }
  ],
  "num_records": 1,
  "_links": {
    "self": {
      "href": "/api/support/ems/events?fields=message.name&max_records=1"
    },
    "next": {
      "href": "/api/support/ems/events?start.keytime=2022-03-
22T10%3A28%3A29-04%3A00&start.node.name=malha-
vsim1&start.index=6260&fields=message.name&max_records=1"
    }
  }
}
```


示例请求：检索包含特定文本和严重性的消息

```
curl -X GET -u admin:<PASSWORD> -k
'https://<IP_ADDRESS>/api/support/ems/events?log_message=*disk*&message.se
verity=alert'
```

响应示例


```
"records": [
    {
      "node": {
        "name": "malha-vsim1",
        "uuid": "da4f9e62-9de3-11ec-976a-005056b369de",
        "_links": {
          "self": {
            "href": "/api/cluster/nodes/da4f9e62-9de3-11ec-976a-
005056b369de"
          }
        }
      },
      "index": 4602,
      "time": "2022-03-18T06:37:46-04:00",
      "message": {
        "severity": "alert",
        "name": "raid.autoPart.disabled"
      },
      "log_message": "raid.autoPart.disabled: Disk auto-partitioning is
disabled on this system: the system needs a minimum of 4 usable internal
hard disks.",
      "_links": {
        "self": {
          "href": "/api/support/ems/events/malha-vsim1/4602"
        }
      }
    }
  ],
  "num_records": 1,
  "_links": {
    "self": {
      "href":
"/api/support/ems/events?log_message=*disk*&message.severity=alert&max_rec
ords=1"
    },
    "next": {
      "href": "/api/support/ems/events?start.keytime=2022-03-
18T06%3A37%3A46-04%3A00&start.node.name=malha-
vsim1&start.index=4602&log_message=*disk*&message.severity=alert"
    }
  }
}
```

## 查看现有 EMS 配置

使用 `get /support/EMS/` API 调用，您可以检索有关现有 EMS 通知配置的信息。

请求示例

```
curl -X GET -u admin:<PASSWORD> -k 'https://<IP_ADDRESS>/api/support/ems/'
```

响应示例

```
{
  "proxy_url": "https://proxyserver.mycompany.com",
  "proxy_user": "proxy_user",
  "mail_server": "mail@mycompany.com",
  "_links": {
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "pubsub_enabled": "1",
  "mail_from": "administrator@mycompany.com"
}
```

## 创建新的 EMS 通知配置

您可以同时使用多个 API 调用来创建新的 EMS 配置，以自定义接收消息的方式和时间。

步骤

1. 配置系统范围的电子邮件设置。例如：

   ```
   curl -X PATCH -u admin:<PASSWORD> -k
   'https://<IP_ADDRESS>/api/support/ems?mail_from=administrator@mycompany.
   com&mail_server=mail@mycompany.com'
   ```

2. 定义使用特定规则与特定消息匹配的筛选器。例如：

   ```
   curl -u admin:<PASSWORD> -X POST -d '{"name": "test-filter",
   "rules.type": ["include"], "rules.message_criteria.severities":
   ["emergency"]}' -k 'https://<IP_ADDRESS>/api/support/ems/filters/'
   ```

3. 为消息创建目标。例如：

```
curl -u admin:<PASSWORD> -X POST -d '{"name": "test-destination",
"type": "email", "destination": "administrator@mycompany.com",
"filters.name": ["important-events"]}' -k
'https://<IP_ADDRESS>/api/support/ems/destinations/'
```

### ONTAP REST API 与 ONTAP 命令行界面命令

与 ONTAP 命令行界面相比， ONTAP REST API 可以使用更少的命令自动执行许多任务。例如，您可以使用一个 POST API 方法来创建筛选器，而不是使用多个 CLI 命令。下表显示了完成常见 EMS 任务与相应的 REST API 调用所需的命令行界面命令：

| ONTAP REST API | ONTAP 命令行界面 |
|---|---|
| `GET /support/EMS` | `event config show` |
| `POST /support/EMS/destinations` | 1. 事件通知目标创建<br>2. 事件通知创建 |
| `GET /support/EMS/Events` | 事件日志显示 |
| `POST /support/EMS/filters` | 1. `event filter create -filter-name <FilterName>`<br>2. `event filter rule add -filter-name <FilterName>` |

### 相关信息

- "ONTAP REST API EMS Python 脚本示例"
- "ONTAP REST API： 自动通知高严重性事件"

# 使用 ONTAP REST API 管理 DACL 和 SACL 文件权限

ONTAP 使用系统访问控制列表（ SACL ）和随机访问控制列表（ DACL ）为文件对象分配权限。从 ONTAP 9.1.1 开始， ONTAP REST API 包括用于为文件分配 SACL 和 DACL 权限以及自动执行文件安全权限的端点。（您可以了解有关在 ONTAP 9.9.1 之前自动执行 SACL 和 DACL 权限的选项的更多信息 "此处"）。

从 ONTAP 9.1.1 开始，您可以使用单个 REST API 调用来代替多个 CLI 命令或 ONTAPI 调用。以下示例显示了如何使用 ONTAP REST API 处理文件权限。在每个示例中，请务必在所示位置用方括号 <> 提供信息值。

您也可以引用示例 "Python 脚本" 该演示了如何自动执行许多与 SACL 和 DACL 相关的活动。

### 查看有效权限

使用 `get /protocols/file-security/effective-permissions/` API 调用，您可以检索特定文件或目

录的当前权限。

请求示例：

```
curl -X GET -u admin:<PASSWORD> -k
'https://<IP_ADDRESS>/api/protocols/file-security/effective-
permissions/cf5f271a-1beb-11ea-8fad-
005056bb645e/administrator/windows/%2F?share.name=sh1&return_records=true'
```

响应示例

```
{
  "svm": {
    "uuid": "cf5f271a-1beb-11ea-8fad-005056bb645e",
    "name": "vs1"
  },
  "user": "administrator",
  "type": "windows",
  "path": "/",
  "share": {
    "path": "/"
  },
  "file_permission": [
    "read",
    "write",
    "append",
    "read_ea",
    "write_ea",
    "execute",
    "delete_child",
    "read_attributes",
    "write_attributes",
    "delete",
    "read_control",
    "write_dac",
    "write_owner",
    "synchronize",
    "system_security"
  ],
  "share_permission": [
    "read",
    "read_ea",
    "execute",
    "read_attributes",
    "read_control",
    "synchronize"
  ]
}
```

## 查看所有审核信息

使用 `get /protocols/file-security/permissions/` API 调用，您可以检索特定文件或目录的所有审核信息。

请求示例:

```
curl -X GET -u admin:<PASSWORD> -k
'https://<IP_ADDRESS>/api/protocols/file-security/permissions/9479099d-
5b9f-11eb-9c4e-0050568e8682/%2Fparent'
```

响应示例


```
{
  "svm": {
    "uuid": "9479099d-5b9f-11eb-9c4e-0050568e8682",
    "name": "vs1"
  },
  "path": "/parent",
  "owner": "BUILTIN\\Administrators",
  "group": "BUILTIN\\Administrators",
  "control_flags": "0x8014",
  "acls": [
    {
      "user": "BUILTIN\\Administrators",
      "access": "access_allow",
      "apply_to": {
        "files": true,
        "sub_folders": true,
        "this_folder": true
      },
      "advanced_rights": {
        "append_data": true,
        "delete": true,
        "delete_child": true,
        "execute_file": true,
        "full_control": true,
        "read_attr": true,
        "read_data": true,
        "read_ea": true,
        "read_perm": true,
        "write_attr": true,
        "write_data": true,
        "write_ea": true,
        "write_owner": true,
        "synchronize": true,
        "write_perm": true
      },
      "access_control": "file_directory"
    },
    {
```

```
      "user": "BUILTIN\\Users",
      "access": "access_allow",
      "apply_to": {
        "files": true,
        "sub_folders": true,
        "this_folder": true
      },
      "advanced_rights": {
        "append_data": true,
        "delete": true,
        "delete_child": true,
        "execute_file": true,
        "full_control": true,
        "read_attr": true,
        "read_data": true,
        "read_ea": true,
        "read_perm": true,
        "write_attr": true,
        "write_data": true,
        "write_ea": true,
        "write_owner": true,
        "synchronize": true,
        "write_perm": true
      },
      "access_control": "file_directory"
    }
  ],
  "inode": 64,
  "security_style": "mixed",
  "effective_style": "ntfs",
  "dos_attributes": "10",
  "text_dos_attr": "----D---",
  "user_id": "0",
  "group_id": "0",
  "mode_bits": 777,
  "text_mode_bits": "rwxrwxrwx"
}
```


## 应用新权限

使用 `POST /protocols/file-security/permissions/` API 调用，您可以将新的安全描述符应用于文件或目录。

请求示例

```
curl -u admin:<PASSWORD> -X POST -d '{ \"acls\": [ { \"access\":
\"access_allow\", \"advanced_rights\": { \"append_data\": true,
\"delete\": true, \"delete_child\": true, \"execute_file\": true,
\"full_control\": true, \"read_attr\": true, \"read_data\": true,
\"read_ea\": true, \"read_perm\": true, \"write_attr\": true,
\"write_data\": true, \"write_ea\": true, \"write_owner\": true,
\"write_perm\": true }, \"apply_to\": { \"files\": true, \"sub_folders\":
true, \"this_folder\": true }, \"user\": \"administrator\" } ],
\"control_flags\": \"32788\", \"group\": \"S-1-5-21-2233347455-2266964949-
1780268902-69700\", \"ignore_paths\": [ \"/parent/child2\" ], \"owner\":
\"S-1-5-21-2233347455-2266964949-1780268902-69304\", \"propagation_mode\":
\"propagate\"}' -k 'https://<IP_ADDRESS>/api/protocols/file-
security/permissions/9479099d-5b9f-11eb-9c4e-
0050568e8682/%2Fparent?return_timeout=0'
```

响应示例

```
{
  "job": {
    "uuid": "3015c294-5bbc-11eb-9c4e-0050568e8682",
    "_links": {
      "self": {
        "href": "/api/cluster/jobs/3015c294-5bbc-11eb-9c4e-0050568e8682"
      }
    }
  }
}
```

## 更新安全描述符信息

使用 `patch /protocols/file-security/permissions/` API 调用，您可以更新文件或目录的特定安全描述符信息，例如主所有者，组或控制标志。

请求示例

```
curl -u admin:<PASSWORD> -X PATCH -d '{ \"control_flags\": \"32788\",
\"group\": \"everyone\", \"owner\": \"user1\"}' -k
'https://<IP_ADDRESS>/api/protocols/file-security/permissions/9479099d-
5b9f-11eb-9c4e-0050568e8682/%2Fparent?return_timeout=0'
```

响应示例

```
{
  "job": {
    "uuid": "6f89e612-5bbd-11eb-9c4e-0050568e8682",
    "_links": {
      "self": {
        "href": "/api/cluster/jobs/6f89e612-5bbd-11eb-9c4e-0050568e8682"
      }
    }
  }
}
```

## 删除现有 SACL/DACL 访问控制条目（ACE）

使用 `Delete /protocols/file-security/permissions/` API 调用，您可以从目录的文件中删除现有 ACE 。此示例将此更改传播到任何子对象。

请求示例

```
curl -u admin:<PASSWORD> -X DELETE -d '{ \"access\": \"access_allow\",
\"apply_to\": { \"files\": true, \"sub_folders\": true, \"this_folder\":
true }, \"ignore_paths\": [ \"/parent/child2\" ], \"propagation_mode\":
\"propagate\"}' -k 'https://<IP_ADDRESS>/api/protocols/file-
security/permissions/9479099d-5b9f-11eb-9c4e-
0050568e8682/%2Fparent/acl/himanshu?return_timeout=0'
```

响应示例

```
{
  "job": {
    "uuid": "e5683b61-5bbf-11eb-9c4e-0050568e8682",
    "_links": {
      "self": {
        "href": "/api/cluster/jobs/e5683b61-5bbf-11eb-9c4e-0050568e8682"
      }
    }
  }
}
```

## ONTAP REST API 与 ONTAP 命令行界面命令

与 ONTAP 命令行界面相比， ONTAP REST API 可以使用更少的命令自动执行许多任务。例如，您可以使用一种 POST API 方法来修改文件的安全描述符，而不是使用多个 CLI 命令。下表显示了完成常见文件系统权限任务与相应的 REST API 调用所需的命令行界面命令：

| ONTAP REST API | ONTAP 命令行界面 |
| --- | --- |
| `get /protocols/file-security/effective-permissions/` | `vserver security file-directory show-effective-permissions` |
| `POST /protocols/file-security/permissions/` | 1. `vserver security file-directory ntfs create`<br>2. `vserver security file-directory ntfs dacl add`<br>3. `vserver security file-directory ntfs sacl add`<br>4. `vserver security file-directory policy create`<br>5. `vserver security file-directory policy task add`<br>6. `Vserver security file-directory apply` |
| `patch /protocols/file-security/permissions/` | `vserver security file-directory ntfs modify` |
| `delete /protocols/file-security/permissions/` | 1. `vserver security file-directory ntfs dacl remove`<br>2. `vserver security file-directory ntfs sacl remove` |

## 相关信息

- "ONTAP REST API DACL/SACL 权限示例 Python 脚本"
- "使用 ONTAP REST API 简化文件安全权限的管理"
- "将专用命令行界面直通与 ONTAP REST API 结合使用（适用于 ONTAP 9.1.1 之前的版本）"

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